

AI Receptivity or AI Adoption Breadth?

A Tool-Specific Reanalysis of the Lower-Literacy/Higher-Usage Link

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(Replication and robustness reanalysis)

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Abstract

Recent evidence reported by [Tully et al. \(2025\)](#) suggests that lower artificial intelligence (AI) literacy predicts greater receptivity toward AI. We revisit this claim using the public data from Study 3 of that article, which measures past usage of five AI tool categories on a five-point frequency scale. We first reproduce the negative association between AI literacy and aggregate AI usage using OLS on participant-level averages, binary logit, ordered logit, and multinomial logit specifications. We then show that the aggregate relationship masks substantial heterogeneity by tool type. In our demographic-adjusted primary specification, AI literacy does not significantly predict text-AI usage (ordered-logit $\beta = -0.090$, $p = .387$), whereas it remains a strong predictor of non-text AI adoption ($\beta = -0.377$, $p < .001$). The non-text effect is also robust under the original Study 3 Table 4 controls ($\beta = -0.502$, $p < .001$). Binary, ordered-logit, and multinomial specifications suggest that the non-text relationship is primarily an adoption/non-adoption pattern rather than evidence of intensive use: the demographic-adjusted odds ratio of ever having used a non-text AI tool is 0.68. These results do not overturn the original finding, but they qualify its interpretation: lower AI literacy may predict broader adoption of certain AI-labelled tools more strongly than it predicts text-AI use.

Keywords: AI literacy, AI adoption, ordinal regression, replication, robustness, construct validity, marketing analytics.

1 Introduction

The proliferation of generative AI products has renewed marketing scholars' interest in the individual-level antecedents of AI adoption ([Puntoni et al., 2021](#); [Hermann and Puntoni, 2024](#)). Among recent contributions, [Tully et al. \(2025\)](#) document a counterintuitive lower-literacy, higher-receptivity link across seven studies: consumers who score lower on objective tests of AI knowledge are, on average, more receptive to AI. Study 3 is especially useful for reanalysis because, unlike Studies 4 and 6, it uses reported past usage of real AI products, rather than stated preferences for hypothetical tasks, as the dependent variable. This makes it a more conservative test of the proposed relationship and, accordingly, a key target for replication and robustness checks.

The original Study 3 (N=401, Amazon Mechanical Turk) measures usage frequency over the previous six months for five AI tool categories: digital image generators (e.g., DALL-E), AI-

powered productivity tools (e.g., Zapier), AI-powered design services (e.g., Canva), AI-powered health apps (e.g., Headspace), and AI-powered writing assistants (e.g., ChatGPT). Each tool is rated on a five-point frequency scale ranging from Never to Weekly. The authors average these five items into an AI receptivity index and regress the index on AI literacy.

We see two reasons to revisit this specification. First, averaging ordinal five-point items into a single continuous index is a standard but contested practice, and the field has accumulated substantial methodological evidence that doing so can distort estimates of treatment effects and even invert their sign (Liddell and Kruschke, 2018; Agresti, 2010). Second, and more substantively, the five tool categories arguably tap heterogeneous behavioral constructs. Text AI (writing assistants) had become highly visible after the release of ChatGPT, whereas image generators, productivity bots, website builders, and meditation apps were plausibly less familiar products. Aggregating them into a single “AI receptivity” index conflates intensive text-AI use with broader adoption of AI-labelled consumer tools.

In this paper we revisit Study 3 along both dimensions. We do not dispute the aggregate lower-literacy/higher-usage association; rather, we show that its substantive meaning changes when AI usage is decomposed by tool type and modelled as ordinal adoption data. The reanalysis makes three points:

1. Replication: the pooled Study 3 association is real. We reproduce the original direction under OLS, binary logit, ordered logit, and multinomial logit.
2. Measurement: averaging ordinal usage responses across heterogeneous AI categories compresses different behavioral processes into a single index.
3. Interpretation: the relationship is much stronger and more robust for non-text AI tools than for text AI alone, and the non-text effect is concentrated at the adoption margin. Study 3 is therefore better interpreted as especially strong evidence about non-text AI adoption breadth rather than as uniform evidence about general AI usage intensity.

2 Background

Algorithm aversion and algorithm appreciation. A long literature in marketing and judgment-and-decision-making documents systematic individual responses to algorithmic agents. Dietvorst et al. (2015) show that people abandon algorithms after observing errors, even when the algorithm outperforms a human alternative. Logg et al. (2019) show the complementary phenomenon, algorithm appreciation, in numerical estimation tasks. Castelo et al. (2019) and Longoni and Cian (2022) qualify both patterns by showing that task type (objective vs. subjective; utilitarian vs. hedonic) moderates consumer responses, while Longoni et al. (2019) document resistance to medical AI driven by uniqueness neglect. Yalcin et al. (2022) show that the direction of the algorithmic decision (favorable vs. unfavorable) matters for consumer reactions, and de Bellis et al. (2023) show that the perceived meaning of manual labor impedes the adoption of autonomous products. Across this literature, contextual and task-related factors play a central role, and individual-level determinants of receptivity remain comparatively understudied.

AI literacy. The construct of AI literacy itself was articulated by Long and Magerko (2020) as a set of competencies needed to evaluate, communicate with, and use AI in everyday contexts. Subsequent reviews (Ng et al., 2021) adapted classical literacy frameworks to characterize AI-specific knowledge, use, evaluation, and ethics. Against this backdrop, Tully et al. (2025) make the novel claim that higher AI literacy decreases receptivity to AI, because demystifying AI dispels perceptions of magic and the awe such perceptions evoke.

Tool-level heterogeneity in AI adoption. Recent macro evidence underscores why aggregating AI tool categories is risky. McElheran et al. (2024) document that, as of 2018, fewer than 6% of U.S. firms used any of five core AI technologies, and adoption was strongly concentrated by industry and firm size. Brynjolfsson et al. (2025) show that productivity gains from generative AI assistants are heterogeneous across workers, with the largest gains accruing to lower-skill agents. Acemoglu and Restrepo (2020) make a parallel point for industrial robots. These results suggest that “AI adoption” is not a single behavioral phenomenon but a family of tool-specific decisions whose drivers may differ across tool categories.

Ordinal versus metric analysis. On the methodological side, McCullagh (1980) introduced the proportional-odds model that we adopt here, and Agresti (2010) provides a comprehensive treatment of ordinal regression. Liddell and Kruschke (2018) review hundreds of psychology articles and document that treating ordinal Likert responses as metric can yield false positives, false negatives, and, in extreme cases, sign inversions of estimated effects. Their recommendation, which we follow, is to estimate ordered-probit or ordered-logit models on the item-level data rather than OLS on averaged indices.

3 Original Study 3 and Analytic Concern

Tully et al. (2025) report that lower AI literacy was associated with greater frequency of AI usage ($B = -0.09$, $SE = .02$, $t(399) = -5.73$, $p < .001$), with the coefficient remaining significant after controlling for technology readiness, general knowledge, motivation for autonomy, and gender ($B = -0.11$, $p < .001$). The dependent variable is the mean of five items $Y_{ij} \in \{1, 2, 3, 4, 5\}$ measuring usage frequency of image generators, productivity tools, website builders, health apps, and writing assistants.

Two features of the design motivate our reanalysis. First, the five items correspond to qualitatively different AI products. Their empirical distributions differ sharply: text AI is used at least occasionally by roughly two-thirds of respondents, while the four non-text categories are heavily concentrated at the lowest category (“Never”), with 65–78% of respondents reporting no usage in the previous six months. Second, the outcome

$$Y_{ij} \in \{1, 2, 3, 4, 5\}$$

is ordinal, sparse for several tools, and repeated within participant. Averaging is not inherently invalid, but it conflates two distinct margins: (i) *adoption* (whether respondents use a tool at

all) and (ii) *intensity* (how often they use it conditional on adoption). When those margins differ across tools, the pooled effect estimate need not have a clean substantive interpretation.

4 Reanalysis Strategy

We download the public Study 3 data from [ResearchBox #1491](#) and use the AI literacy score (SC0, summed correct answers on the 17-item AI-constructed measure) and the five usage items. Our primary specification is demographic-adjusted: age, income, general knowledge, motivation for autonomy, and a male-gender indicator. We use this specification because it retains the full Study 3 analytic sample (N=401) while adjusting for demographics and the two non-technology individual-difference covariates emphasized in the original paper. As a robustness check, we re-estimate the key models using the original Table 4 control set: technology readiness, general knowledge, motivation for autonomy, and a male-gender indicator (N=379 because technology readiness has missing observations). We standardize the AI literacy score and continuous covariates so that coefficients are interpretable as the effect of a one-standard-deviation increase. We then fit four model families across three outcome groupings.

Model families.

1. OLS on the participant-level average. This is the closest match to the original Study 3 specification and serves as our pooled benchmark.
2. Binary logit on item-level data with task fixed effects. We consider two thresholds: $Y > 3$ (“frequent use”) and $Y > 1$ (“adoption”).
3. Ordered logit (proportional-odds model) ([McCullagh, 1980](#)). This is the natural model for an ordered Likert response with $K = 5$ categories and is robust to violation of metric-scale assumptions ([Liddell and Kruschke, 2018](#)).
4. Multinomial logit, treating the five categories as nominal. This is a stress test: it does not impose proportional odds or any ordering and is therefore robust to violations of either.

Outcome groupings.

1. Pooled (5 tools): the original Study 3 outcome.
2. Text only: `AI_text` alone.
3. Non-text only: `AI_image`, `AI_productivity`, `AI_website`, `AI_healthapp`.

Implementation. All models are fit in Python with `statsmodels`. OLS and binary-logit standard errors are heteroskedasticity-robust (HC3). Item-level models include task fixed effects. Full code, a Jupyter notebook, and pre-generated figures are provided as supplementary material.

Table 1: Demographic-adjusted text vs. non-text decomposition. Coefficient on a +1 standard-deviation increase in AI literacy, controlling for age, income, general knowledge, motivation for autonomy, and male gender. Item-level models include task fixed effects.

Outcome / Model	Coef.	SE	p	Notes
Text AI only (N=401)				
OLS	-0.077	0.082	.351	HC3 SE
Binary logit ($Y > 3$)	-0.103	0.141	.467	OR = 0.90
Binary adoption ($Y > 1$)	-0.061	0.128	.634	OR = 0.94
Ordered logit	-0.090	0.104	.387	Prop. odds
Non-text AI (4 tools; 1,604 item-level obs.)				
OLS on participant avg.	-0.207	0.054	< .001	HC3 SE
Binary logit ($Y > 3$)	-0.398	0.103	< .001	OR = 0.67
Binary adoption ($Y > 1$)	-0.388	0.069	< .001	OR = 0.68
Ordered logit	-0.377	0.063	< .001	Prop. odds

5 Results

5.1 Replication of the pooled effect

The demographic-adjusted pooled five-tool association is negative and statistically significant in every specification. OLS on the participant-level average gives $\hat{\beta} = -0.181$ ($p = .001$), after standardization. The ordered logit gives $\hat{\beta} = -0.307$ ($p < .001$), the binary logit at the scale midpoint gives $\hat{\beta} = -0.320$ ($p < .001$, odds ratio = 0.73), and the binary adoption logit gives $\hat{\beta} = -0.330$ ($p < .001$, odds ratio = 0.72). The multinomial logit yields the same negative direction across all four non-reference categories. We therefore reproduce the original pooled lower-literacy/higher-usage association before decomposing it by tool type.

5.2 Text vs. non-text decomposition

Table 1 re-estimates the ordered-logit and binary-logit specifications separately on the text-AI outcome and on the four non-text outcomes. The demographic-adjusted contrast is clear. For text AI alone, the ordered-logit coefficient is small and not significantly different from zero ($\hat{\beta} = -0.090$, $SE = 0.104$, $p = .387$); the binary “frequent use” logit and the binary adoption logit are also non-significant. For non-text AI, the ordered-logit coefficient is roughly four times larger in magnitude and highly significant ($\hat{\beta} = -0.377$, $SE = 0.063$, $p < .001$), and the result is consistent across the ordered, midpoint-binary, and adoption-binary specifications.

The contrast is even sharper at the adoption margin. The binary adoption specification ($Y > 1$, i.e., “ever used in the last six months”) gives an odds ratio of 0.68 per one-standard-deviation increase in AI literacy for non-text tools, a clean, easily interpretable result, and an odds ratio almost indistinguishable from 1.0 for text AI.

Figure 1 visualizes the consequence of this asymmetry. We plot the ordered-logit-predicted probability $\Pr(Y = 1)$ (“Never used”) over the empirical range of standardized literacy, separately for the text and non-text models. For text AI, the predicted probability of non-use rises only modestly, from 0.27 at $z = -2$ to 0.35 at $z = +2$. For non-text AI, it rises sharply, from 0.50 to

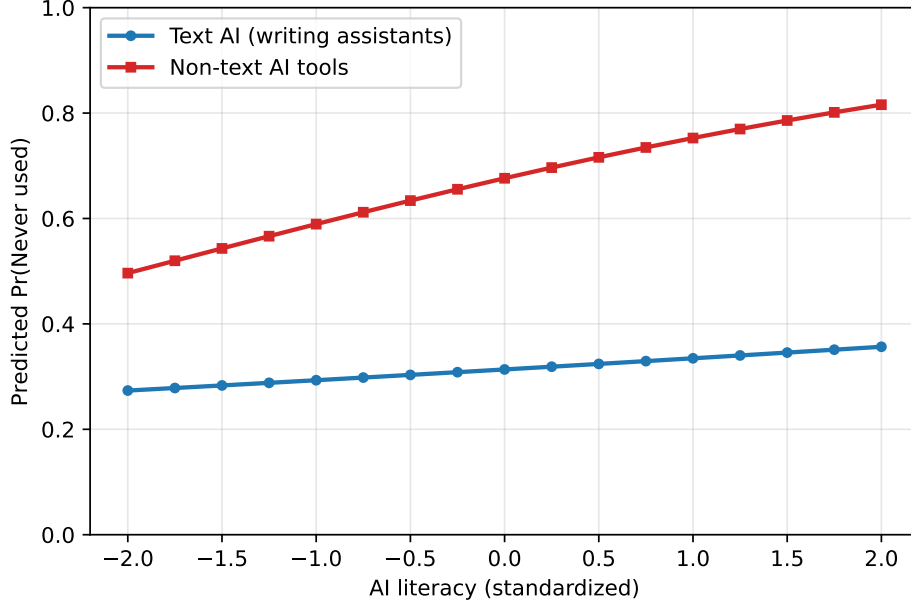


Figure 1: Predicted probability of non-use by AI literacy. Curves are derived from ordered-logit fits on text-AI only (blue) and on the four non-text AI tools (red, with task fixed effects), holding all covariates at their sample means. Higher AI literacy predicts much higher odds of having never used non-text AI tools; the relationship is roughly flat for text AI.

0.82 across the same range. In other words, the higher a respondent’s AI literacy, the more likely they are to have never tried image generators, productivity bots, website builders, or health apps, while their probability of having used a ChatGPT-style writing assistant is essentially flat.

5.3 Original Table 4 covariate check

Table 2 repeats the decomposition using the original Study 3 Table 4 controls: technology readiness, general knowledge, motivation for autonomy, and male gender. Because technology readiness has missing observations, this specification uses $N=379$. The non-text pattern strengthens: the ordered-logit coefficient is -0.502 and the adoption odds ratio is 0.61. The text-AI coefficient also becomes negative and statistically significant in the ordered and midpoint-binary models, although not at the adoption margin. We therefore avoid interpreting the text-AI result as a true zero. The more stable conclusion is comparative: the literacy gradient is larger and more consistently adoption-based for non-text AI tools.

6 Discussion

The original aggregate association reported by [Tully et al. \(2025\)](#) is robust: it survives every reasonable parametric alternative to OLS on an averaged Likert index. What changes when the outcome is decomposed is its substantive interpretation. The pooled estimate combines three analytically distinct constructs:

1. General AI receptivity: a latent willingness to adopt AI across product categories.

Table 2: Robustness check using the original Study 3 Table 4 covariates. Coefficient on a +1 standard-deviation increase in AI literacy, controlling for technology readiness, general knowledge, motivation for autonomy, and male gender. Item-level models include task fixed effects.

Outcome / Model	Coef.	SE	<i>p</i>	Notes
Text AI only (N=379)				
Binary logit ($Y > 3$)	−0.322	0.154	.037	OR = 0.72
Binary adoption ($Y > 1$)	−0.238	0.145	.100	OR = 0.79
Ordered logit	−0.290	0.112	.010	Prop. odds
Non-text AI (4 tools; 1,516 item-level obs.)				
Binary logit ($Y > 3$)	−0.577	0.112	< .001	OR = 0.56
Binary adoption ($Y > 1$)	−0.497	0.075	< .001	OR = 0.61
Ordered logit	−0.502	0.067	< .001	Prop. odds

2. Text-AI usage intensity: how often a respondent uses text-based generative AI tools such as ChatGPT, conditional on having adopted them.
3. Non-text AI adoption breadth: the probability of having tried at least one of several heterogeneous AI-labelled consumer products.

Our results suggest that Study 3 speaks most clearly to the third construct. The text-AI margin is weaker and less stable than the non-text margin: it is small and non-significant in the full-sample demographic-adjusted specification, but becomes negative and statistically significant in the original Table 4 control specification. By contrast, the non-text relationship remains large across specifications and appears at both the ordered-response and adoption margins. The aggregate estimate is therefore carrying substantial information about adoption breadth across categories where the modal respondent has never used the product.

This re-interpretation does not contradict the broader theoretical claim that lower-literacy consumers may perceive AI as magical (Tully et al., 2025). But it does change what that claim implies for managers. The original article suggests that firms may benefit from targeting lower-AI-literacy consumers. Our results narrow this prescription in two ways. First, the targeting case is much stronger for low-penetration, non-text AI product categories (image generation, productivity automation, AI design services, AI wellness apps) than for more familiar text assistants. Second, the non-text effect is driven primarily by the adoption margin rather than by intensive use: low-literacy consumers are differentially more likely to try these products, not necessarily to use them more intensively conditional on trial. This matters for product strategy. Lifetime-value calculations that assume a uniform usage-intensity response to literacy may overestimate the value of low-literacy acquisitions.

Methodologically, our reanalysis illustrates the value of treating heterogeneous ordinal items as what they are. Averaging five items with sharply different empirical distributions into a single “receptivity” index is convenient, and in this case it does not invert any sign, but it does flatten a multi-tool adoption pattern into something that reads like a single dispositional trait. The construct-validity question that follows, whether “AI receptivity” is one thing or several, deserves attention beyond the present application (Long and Magerko, 2020; Ng et al., 2021).

7 Limitations

Several limitations of our reanalysis warrant explicit mention.

- No new data. We rely entirely on the data and codebook released by the original authors. Their Studies 1, 2, and 4–7 use different measures of receptivity (cross-country adoption readiness, propensity to use generative AI for assignments, and relative preference for AI vs. human task completion) for which our critique does not directly apply.
- Observational and self-reported. Like the original Study 3, our analyses are correlational and rely on self-reported usage. Any measurement error in self-reported usage that correlates with literacy could bias the estimated coefficients.
- Post hoc analysis. Our decomposition is not preregistered. We have nevertheless implemented the four model families exactly as described in our analytic strategy and provide all code.
- No causal claim. We make no claim that AI literacy causes reduced adoption of any tool. The mediation analysis in the original paper (perceptions of AI as magical) is not addressed here and remains the authors’ primary theoretical mechanism.
- No test of magicalness by tool type. An obvious extension would be to ask whether perceived “magicalness” varies systematically across non-text and text tool categories, and whether that variation can account for the asymmetry we document. This would require new data.
- Standard errors. OLS and binary-logit standard errors are heteroskedasticity-robust, but the ordered-logit estimates use the standard maximum-likelihood covariance matrix available in `statsmodels`. A mixed-effects or cluster-robust ordinal model would be a useful additional robustness check.

8 Conclusion

We do not dispute the aggregate lower-literacy/higher-usage association reported by [Tully et al. \(2025\)](#); rather, we show that its substantive meaning changes when AI usage is decomposed by tool type and modelled as ordinal adoption data. The negative association replicates, but it is stronger and more stable for non-text AI categories, where it operates primarily at the adoption margin rather than at the intensity margin. We therefore read Study 3 as especially strong evidence about non-text AI adoption breadth, with weaker and specification-sensitive evidence about text-AI use. This qualification preserves the spirit of the original contribution while sharpening its managerial and theoretical implications.

Data and code availability

The Study 3 data are publicly available from the original authors at [ResearchBox #1491](#). All Python code used to produce the tables and figures in this paper, together with a fully executed

Jupyter notebook, is available in the project repository: [HristoInouzhe/AI-use-vs-AI-literacy](#).

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